Solutions for Digital Age

Training Course Content

Basic Training Course Content

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LINUX FOUNDATIONS

Shell basics

- Navigating the command line and the file system
- Command structure, arguments and options
- Input and output redirection and pipes
- Environment variables and shell configuration files

★ **Practical Task: Complete a set of guided shell exercises covering navigation and redirection**

File system hierarchy

- Standard directory layout and the purpose of key directories
- Absolute and relative paths
- Mounting and the role of the file system table
- Locating files and configuration

Permissions deep dive

- The read, write and execute model
- Ownership and group membership
- Octal and symbolic notation
- Special bits: setuid, setgid and the sticky bit
- The umask and default permissions

★ **Practical Task: Correct a set of broken permission scenarios on a practice host**

Users, groups and sudo

- Creating and managing users and groups
- Password and account aging settings
- The sudo model and the sudoers file
- Switching users safely

Text processing with grep, sed, awk and pipes

- Searching text with grep
- Stream editing with sed
- Field based processing with awk
- Combining tools through pipes

★ **Practical Task: Extract and summarise data from a sample log file**

LINUX OPERATIONS

Software and package management

- Package managers and repositories
- Installing, updating and removing packages
- Querying installed software
- Managing repository sources

Services and systemd

- The service concept
- Unit files and their structure
- Starting, stopping and enabling services
- Reading service status

Process management

- Viewing running processes
- Foreground and background jobs
- Signals and terminating processes
- Priorities and resource usage

Logs with journalctl and the log directory

- The system journal
- Filtering and following log output
- Traditional log files
- Log rotation basics

Cron and timers

- Scheduling work with cron
- Crontab syntax
- Systemd timers
- Verifying scheduled jobs

Storage basics

- Disks and partitions
- File systems and mounting
- Checking disk and inode usage
- Persistent mounts

SSH and key based authentication

- The SSH client and server
- Generating and distributing keys
- The SSH client configuration file
- Copying files with scp and rsync

Shell scripting intro

- Variables and quoting
- Conditionals and loops
- Exit codes
- Writing a simple reusable script

Install and configure Apache or Nginx

- Installing through the package manager
- Working with the main configuration files
- Serving a test site
- Starting and verifying the service

  ★ **Practical Task: Publish a sample page and confirm it is reachable**

Harden SSH

- Disabling root login
- Enforcing key based login
- Restricting which users may connect
- Limiting authentication attempts and idle sessions

Schedule a backup script

- Writing a backup script
- Scheduling it with cron or a timer
- Verifying output and logs

  ★ **Practical Task: Build and schedule a working backup job**

RAID

- The RAID concept and its purpose
- Common RAID levels
- An overview of software RAID
- Why RAID is not a backup

NETWORKING FUNDAMENTALS

OSI and TCP and IP models

- The layered model
- The role of each layer
- Encapsulation
- Mapping protocols to layers

IP addressing and subnetting

- IPv4 address structure
- Subnet masks and CIDR notation
- Calculating networks and host counts
- Public and private address ranges

  ★ **Practical Task: Solve a set of subnetting problems**

Routing and switching concepts

- Switching at Layer 2
- Routing at Layer 3
- The default gateway
- The routing table

Static routes and gateways

- Adding and removing routes
- Configuring a default gateway
- Inspecting routes from the command line

DNS with dig, nslookup and the resolution flow

- The resolution process
- Common record types
- Querying with dig and nslookup
- Local resolver configuration

Diagnostic tools: ping, traceroute, ss and netstat

- Testing reachability
- Tracing a network path
- Viewing sockets and listening ports
- Interpreting the results

HTTP, HTTPS and TLS at a high level

- The request and response cycle
- Common status codes
- The role of TLS
- The handshake at a high level

SSL certificate basics

- What a certificate contains
- Certificate authorities and trust chains
- Validity periods and expiry
- Viewing a certificate

NETWORK OPERATIONS AND SECURITY

VLANs: access and trunk ports with 802.1Q

- The purpose of VLANs
- Access ports
- Trunk ports and 802.1Q tagging
- The native VLAN

NAT concepts

- Why NAT is used
- Source and destination NAT
- Port address translation
- NAT and connectivity

iptables: chains, tables, filter and nat

- Tables and chains
- Rule structure and targets
- The filter table
- The nat table

Packet analysis with Wireshark and tcpdump

- Capturing traffic
- Applying capture and display filters
- Reading a capture
- Command line capture with tcpdump

  ★ Practical Task: Capture and analyse traffic from a simple exchange

Firewall with ufw, firewalld and apparmor

- Simplified rules with ufw
- Zone based control with firewalld
- Application confinement with apparmor
- Choosing the right tool

DOCKER FUNDAMENTALS

Container and virtual machine concepts

- How containers differ from virtual machines
- Isolation and the shared kernel
- When to use each

Docker architecture and installation

- The Docker engine and client
- Images, registries and containers
- Installing Docker

Images and containers

- Pulling and listing images
- Creating containers from images
- The image and container relationship

Docker Hub and registries

- Public and private registries
- Searching and pulling images
- Tags and versions

Container lifecycle: run, stop, inspect, logs and exec

- Running and stopping containers
- Inspecting container configuration
- Viewing container logs
- Executing commands inside a container

DOCKER IN PRACTICE

Volumes and bind mounts

- Persisting data with volumes
- Bind mounting host paths
- Volume drivers

Networking in Docker

- Bridge networks
- Host and none networks
- User defined networks and DNS

Docker Compose

- Compose file structure
- Defining services
- Managing multi container applications

Building images with Dockerfiles

- Dockerfile instructions
- Layering and caching
- Optimising image size

Best practices

- Least privilege in containers
- Image scanning
- Managing secrets

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- Pods, deployments and services
- The control plane
- The node and kubelet

Installing and using kubectl

- Cluster access and kubeconfig
- Common commands
- Inspecting cluster resources

Deployments and scaling

- Creating deployments
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- Rollbacks

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- ClusterIP, NodePort and LoadBalancer
- DNS inside the cluster
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- Externalising configuration
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SECURITY ACROSS THE STACK

Linux hardening basics

- Reducing the attack surface
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- CIS Benchmark awareness

SSH hardening

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- Disabling weak ciphers
- Fail2ban and rate limiting

Application security basics

- Input validation
- Least privilege for service accounts
- Secrets handling in scripts and config

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- Running as non root
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- Capability dropping

MONITORING

Prometheus and Grafana

- Metrics collection with Prometheus
- Visualising data in Grafana
- Alerting rules

Log aggregation

- Centralising logs
- Searching and filtering
- Retention policies

Health checks and uptime

- Endpoint health checks
- Uptime monitoring
- On call basics